Part 2: Applying OLTP to OLAP Transformation

Task 1: MySQL OLTP Schema (Transactional)

File Name: Part 2_Task 1.sql

This script builds the live operational database in MySQL.

SQL

```sql
-- PART 2 TASK 1: OLTP SCHEMA (MySQL)
CREATE TABLE Roles (RoleID INT PRIMARY KEY, RoleName VARCHAR(50));
CREATE TABLE Patients (PatientID INT PRIMARY KEY, FullName VARCHAR(100), DOB DATE,
Gender VARCHAR(10));
CREATE TABLE Employees (EmployeeID INT PRIMARY KEY, FullName VARCHAR(100), RoleID
INT, FOREIGN KEY (RoleID) REFERENCES Roles(RoleID));
CREATE TABLE Appointments (AppointmentID INT PRIMARY KEY, PatientID INT, EmployeeID INT,
AppointmentDate DATETIME, Status VARCHAR(20), FOREIGN KEY (PatientID) REFERENCES
Patients(PatientID), FOREIGN KEY (EmployeeID) REFERENCES Employees(EmployeeID));
CREATE TABLE Billing (BillID INT PRIMARY KEY, AppointmentID INT, Amount DECIMAL(10,2),
Status VARCHAR(20), FOREIGN KEY (AppointmentID) REFERENCES
Appointments(AppointmentID));

-- SEED DATA
INSERT INTO Roles VALUES (1, 'Doctor'), (2, 'Specialist');
INSERT INTO Patients VALUES (101, 'John Doe', '1985-06-15', 'Male');
INSERT INTO Employees VALUES (501, 'Dr. Smith', 1);
INSERT INTO Appointments VALUES (2001, 101, 501, '2026-02-12 10:30:00', 'Completed');
INSERT INTO Billing VALUES (9001, 2001, 350.00, 'Paid');
```